

Categories, Proofs, and Processes: Mini-Project

1081502

Hilary Term

1

a

Definition 1.1 ([1]). A natural isomorphism is a natural transformation $t : F \rightarrow G$ where each arrow t_A in its family of arrows is isomorphic:

$$t_A : F A \xrightarrow{\cong} G A$$

▲

Definition 1.2 ([6], [7] [2]). A bicategory B consists of

- A collection of objects $a, b, c, \dots$ called **0-cells**.
- For each pair of 0-cells, a, b , a category $B(a, b)$ where its
 - Objects are morphisms between a, b called **1-cells**.
 - Morphisms between objects are called **2-cells**, or equivalently, natural transformations.

$$\begin{array}{ccc} & f & \\ & \longrightarrow & \\ a & \Downarrow \eta & b \\ & \longrightarrow & \\ & g & \end{array}$$

- For each 0-cell a , the 1-cell $1_a \in B(a, a)$ called the identity cell at a .
- For each triple of 0-cells, a, b, c , a bifunctor

$$\circ : B(b, c) \times B(a, b) \rightarrow B(a, c)$$

called **horizontal composition**.

- For each 1-cell $f : a \rightarrow b$, the 2-cell identity $1_f : f \Rightarrow f$ called the 2-identity.
- For each triple of 1-cells, $f, g, h : a \rightarrow b$ with 2-cells $\eta : f \Rightarrow g, \theta : g \Rightarrow h$, a vertical composite

$$\theta \bullet \eta : f \Rightarrow h$$

$$\begin{array}{ccc} & f & \\ & \longrightarrow & \\ a & \Downarrow \eta & b \\ & \Downarrow \theta & \\ & \longrightarrow & \\ & h & \end{array} \qquad \begin{array}{ccc} & f & \\ & \longrightarrow & \\ a & \Downarrow \theta \bullet \eta & b \\ & \longrightarrow & \\ & h & \end{array}$$

- For each triple of 0-cells a, b, c , 1-cells $f, g : a \rightarrow b$, $h : b \rightarrow c$, 2-cell $\eta : f \Rightarrow g$, a **left whiskering**

$$h \triangleleft \eta : h \circ f \Rightarrow h \circ g$$

- For each triple of 0-cells a, b, c , 1-cells $f : a \rightarrow b$, $g, h : b \rightarrow c$, 2-cell $\eta : g \Rightarrow h$, a **right whiskering**

$$\eta \triangleright f : g \circ f \Rightarrow h \circ f$$

- Natural isomorphisms ρ, λ where for each 1-cell $f : a \rightarrow b$, components ρ_f, λ_f called **right and left unitors** respectively where

$$\rho_f : f \circ 1_a \Rightarrow f, \quad \lambda_f : 1_b \circ f \Rightarrow f$$

- A natural isomorphism α , where for each triple of 1-cells $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$, the component $\alpha_{h,g,f}$ called an **associator** defined as

$$\alpha_{h,g,f} : h \circ (g \circ f) \Rightarrow (h \circ g) \circ f$$

such that the following conditions hold

- The *pentagon identity* is satisfied by the associators - for any quadruple of 1-cells

$$a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d \xrightarrow{k} e$$

the following pentagon must commute.

$$\begin{array}{ccc} k \circ (h \circ (g \circ f)) & \xrightarrow{\alpha_{k,h,g \circ f}} & (k \circ h) \circ (g \circ f) \xrightarrow{\alpha_{k \circ h, g, f}} ((k \circ h) \circ g) \circ f \\ \downarrow 1_k \triangleleft \alpha_{h,g,f} & & \downarrow \alpha_{k,h,g} \triangleright 1_f \\ k \circ ((h \circ g) \circ f) & \xrightarrow{\alpha_{k,h \circ g, f}} & (k \circ (h \circ g)) \circ f \end{array}$$

- The *triangle identity* is satisfied by the unitors - for any pair of 1-cells $a \xrightarrow{f} b \xrightarrow{g} c$, the following triangle must commute.

$$\begin{array}{ccc} g \circ (1_b \circ f) & & \\ \alpha_{g,1_b,f} \downarrow & \searrow 1_g \triangleleft \lambda_{a,b} & \\ (g \circ 1_b) \circ f & \xrightarrow{\rho_{a,b} \triangleright 1_f} & g \circ f \end{array}$$

A bicategory is also called a *weak 2-category*. ▲

b

Definition 1.3 (Monoidal Category [1]). A monoidal category is a structure $(\mathcal{C}, \otimes, I, \alpha, \lambda, \rho)$ where

- $\mathcal{C}$ is a category,
- $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ is a functor (*monoidal product*),
- I is a distinguished object of $\mathcal{C}$ (*unit*)
- α, λ, ρ are natural isomorphisms (*associator, left unitor, right unitor*) with components

$$\alpha_{A,B,C} : A \otimes (B \otimes C) \xrightarrow{\cong} (A \otimes B) \otimes C$$

$$\lambda_A : I \otimes A \xrightarrow{\cong} A, \quad \rho_A : A \otimes I \xrightarrow{\cong} A$$

such that $\lambda_I = \rho_I : I \otimes I \rightarrow I$ and the following diagrams commute.

$$\begin{array}{ccc} A \otimes (B \otimes (C \otimes D)) & \xrightarrow{\alpha_{A,B,C \otimes D}} (A \otimes B) \otimes (C \otimes D) & \xrightarrow{\alpha_{A \otimes B,C,D}} ((A \otimes B) \otimes C) \otimes D \\ \downarrow 1_A \otimes \alpha_{B,C,D} & & \downarrow \alpha_{A,B,C} \otimes 1_D \\ A \otimes ((B \otimes C) \otimes D) & \xrightarrow{\alpha_{A,B \otimes C,D}} & (A \otimes (B \otimes C)) \otimes D \end{array}$$

$$\begin{array}{ccc} A \otimes (I \otimes B) & & \\ \downarrow \alpha_{A,I,B} & \searrow 1_A \otimes \lambda_B & \\ (A \otimes I) \otimes B & \xrightarrow{\rho_A \otimes 1_B} & A \otimes B \end{array}$$

▲

Definition 1.4 ([6], [7]). A monoidal category M is a bicategory BM with a single object $\bullet$ where

- Objects A of M become 1-cells $[A] : \bullet \rightarrow \bullet$ of BM
- Composition of 1-cells in BM follow the definition

$$[A] \circ [B] := [A \otimes B]$$

using the monoidal product $\otimes$ of M .

- The monoidal unit I in M becomes the identity cell $1_\bullet := [I] : \bullet \rightarrow \bullet$ in BM .
- Morphisms $f : A \rightarrow B$ in M become 2-cells $[f] : [A] \Rightarrow [B]$ in BM .
- Identity morphisms 1_A in M become 2-identities $[1_A]$ in BM .
- Natural isomorphism α of M where

$$\alpha_{A,B,C} : (A \otimes B) \otimes C \Rightarrow A \otimes (B \otimes C)$$

becomes the associator

$$\alpha_{[A],[B],[C]} : ([A] \circ [B]) \circ [C] \Rightarrow [A] \circ ([B] \circ [C])$$

for the natural isomorphism α of BM

- Natural isomorphisms λ, ρ where each component

$$\lambda_A : I \otimes A \Rightarrow A, \quad \rho_A : A \otimes I \Rightarrow A$$

of M respectively becomes the left and right unitors

$$\lambda_{[A]} : [I] \circ [A] \Rightarrow [A], \quad \rho_{[A]} : [A] \circ [I] \Rightarrow [A]$$

for the natural isomorphisms λ, ρ of BM .

▲

The coherence conditions then hold, where the following diagrams commute in BM iff the corresponding diagrams [1] commute in M .

- Unitors

$$\begin{array}{ccc} [A] \circ ([I] \circ [B]) & & A \otimes (I \otimes B) \\ \alpha_{[B],[I],[A]} \downarrow & \searrow 1_{[A]} \triangleleft \lambda_{[B]} & \alpha_{A,I,B} \downarrow \quad \searrow 1_A \otimes \lambda_B \\ ([A] \circ [I]) \circ [B] & \xrightarrow{\rho_{[A]} \triangleright 1_{[B]}} [A] \circ [B] & (A \otimes I) \otimes B \xrightarrow{\rho_A \otimes 1_B} A \otimes B \end{array} \quad \Longleftrightarrow$$

- Associators

$$\begin{array}{ccc} [A] \circ ([B] \circ ([C] \circ [D])) & \xrightarrow{\alpha_{[A],[B],[C] \circ [D]}} ([A] \circ [B]) \circ ([C] \circ [D]) & \xrightarrow{\alpha_{[A] \circ [B],[C],[D]}} (([A] \circ [B]) \circ [C]) \circ [D] \\ 1_{[A]} \triangleleft \alpha_{[B],[C],[D]} \downarrow & & \downarrow \alpha_{[A],[B],[C] \triangleright 1_{[D]}} \\ [A] \circ (([B] \circ [C]) \circ [D]) & \xrightarrow{\alpha_{[A],[B] \circ [C],[D]}} & ([A] \circ ([B] \circ [C])) \circ [D] \end{array}$$

$\Longleftrightarrow$

$$\begin{array}{ccc} A \otimes (B \otimes (C \otimes D)) & \xrightarrow{\alpha_{A,B,C \otimes D}} (A \otimes B) \otimes (C \otimes D) & \xrightarrow{\alpha_{A \otimes B,C,D}} ((A \otimes B) \otimes C) \otimes D \\ 1_A \otimes \alpha_{B,C,D} \downarrow & & \downarrow \alpha_{A,B,C \otimes 1_D} \\ A \otimes ((B \otimes C) \otimes D) & \xrightarrow{\alpha_{A,B \otimes C,D}} & (A \otimes (B \otimes C)) \otimes D \end{array}$$

2

a ... , a monoid is a category, ...

First, we must list the definitions of both a monoid and category.

Definition 2.1 (Monoid [1]). A **monoid** is a structure $(M, \cdot, 1)$ where M is a set,

$$- \cdot - : M \times M \rightarrow M$$

is a binary operation, and $1 \in M$, satisfying the axioms $\forall x, y, z \in M$.

$$(x \cdot y) \cdot z = x \cdot (y \cdot z), \quad 1 \cdot x = x = x \cdot 1$$

▲

Definition 2.2 (Category [1] [8] [3]). A **category** consists of

- A collection $\text{Ob}(\mathcal{C})$ of **objects**.
- A collection $\text{Ar}(\mathcal{C})$ of **arrows** or morphisms.
- For every arrow f , an object $s(f)$ called its source or domain, and an object $t(f)$ called its target or codomain. An arrow with domain $s(f) = A$ and codomain $t(f) = B$ is denoted $f : A \rightarrow B$.
- For each pair of objects A, B , we define the set

$$\mathcal{C}(A, B) := \{f \in \text{Ar}(\mathcal{C}) \mid f : A \rightarrow B\}$$

called the **hom-set**.

- For each pair of morphisms f in $\mathcal{C}(A, B)$, and g in $\mathcal{C}(B, C)$, a composite morphism $g \circ f$ in $\mathcal{C}(A, C)$.

$$A \xrightarrow{f} B \xrightarrow{g} C$$

- For each object A , an **identity** arrow $1_A : A \rightarrow A$

where the following axioms are satisfied whenever the domains and codomains match so that compositions are well-defined.

- Associativity

$$\forall f, g, h \in \text{Ar}(\mathcal{C}), h \circ (g \circ f) = (h \circ g) \circ f$$

- Unit

$$\forall f : A \rightarrow B, f \circ 1_A = f = 1_B \circ f$$

▲

A monoid $(M, \cdot, 1_M)$ can then be defined as a category $\mathcal{M}$ with one object $*$ where

- Each arrow $f_e : * \rightarrow *$ of the category corresponds to each element $e \in M$ of the monoid.
- The source and target of all arrows are the same object $*$.
- There is a single hom-set $\mathcal{M}(*, *)$.
- Composition $f_b \circ f_a$ corresponds to applying binary operations $- \cdot a \cdot b$
- The identity arrow on $*$, 1_* corresponds to the unit of the monoid $1_M \in M$.

so the axioms are satisfied in the category $\mathcal{M}$ iff the axioms are satisfied in the monoid $(M, \cdot, 1_M)$.

- Associativity

$$\forall f_a, f_b, f_c \in \text{Ar}(\mathcal{M}), f_c \circ (f_b \circ f_a) = (f_c \circ f_b) \circ f_a \iff \forall a, b, c \in M, (a \cdot b) \cdot c = a \cdot (b \cdot c)$$

- Unit

$$\forall f_a \in \text{Ar}(\mathcal{M}), f_a \circ 1_* = f_a = 1_* \circ f_a \iff \forall a \in M, 1_M \cdot a = a = a \cdot 1_M$$

So every monoid is a special kind of category, and categories are the more general concept.

b ... , a category is a monad, ...

We note the following definitions.

Definition 2.3 (Strict). A strict bicategory is a bicategory where the natural isomorphisms α, ρ, λ are identities. ▲

Definition 2.4 (Strict 2-category [5]). Strict 2-categories are strict bicategories. Conversely, bicategories are weak 2-categories. ▲

We can note that bicategories are then the generalisation of strict 2-categories, the notion of a *general 2-category*.

One way to define the concept of monads is the following.

Note 2.5.

$$T^2 := T \circ T, \quad T^3 := T \circ T^2$$

Definition 2.6 (Monad [1] [10]). A monad over a category $\mathcal{C}$ is a triple (T, η, μ) where $T : \mathcal{C} \rightarrow \mathcal{C}$ is an endofunctor and $\eta : 1_{\mathcal{C}} \rightarrow T, \mu : T^2 \rightarrow T$ are natural transformations such that the following diagram commutes.

$$\begin{array}{ccc} T^3 A & \xrightarrow{\mu_{TA}} & T^2 A \\ T\mu_A \downarrow & & \downarrow \mu_A \\ T^2 A & \xrightarrow{\mu_A} & TA \end{array} \quad \begin{array}{ccc} TA & \xrightarrow{\eta_{TA}} & T^2 A \\ T\eta_A \downarrow & \searrow 1_{TA} & \downarrow \mu_A \\ T^2 A & \xrightarrow{\mu_A} & TA \end{array}$$

η is the unit, and μ is the multiplication of the monad. ▲

This definition, however, can easily be generalised using 2-categories.

Definition 2.7 (Monad [11] [10]). Let $\mathcal{C}$ denote any bicategory, strict or otherwise. A monad (X, S) consists of

- A 0-cell X
- A 1-cell $S : X \rightarrow X$
- A pair of 2-cells $\eta : 1_S \rightarrow S, \mu : SS \rightarrow S$ called the *unit* and *multiplication* of the monad.

such that the following diagrams commute, up to a natural isomorphism.

$$\begin{array}{ccc} S^3 & \xrightarrow{\mu_S} & S^2 \\ S\mu \downarrow & & \downarrow \mu \\ S^2 & \xrightarrow{\mu} & S \end{array} \quad \begin{array}{ccc} S & \xrightarrow{\eta_S} & S^2 \\ S\eta \downarrow & \searrow 1_S & \downarrow \mu \\ S^2 & \xrightarrow{\mu} & S \end{array}$$

where the following notation is used.

$$S^2 := S \circ S := SS, \quad S^3 := S \circ S^2 := SSS$$

$$S\mu := S \triangleleft \mu, \quad \mu S := \mu \triangleright S$$

▲

Definition 2.8. If we let the strict 2-category **Cat** of categories, functors, and natural transformations be the bicategory in Def. 2.7, then we get Def. 2.6, where

- The 0-cell is the category $\mathcal{C}$ in Def. 2.6
- The 1-cell is the endofunctor on $T : \mathcal{C} \rightarrow \mathcal{C}$
- The 2-cells are the natural transformations $\mu : T^2 \rightarrow T, \eta : 1_T \rightarrow T$.

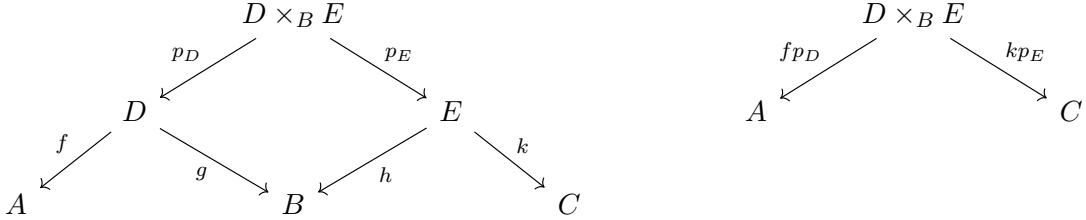
▲

Definition 2.9 (Span [9] [10]). A span in a category $\mathcal{C}$ from an object A to B is a diagram of the form

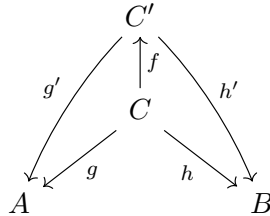
$$A \xleftarrow{f} C \xrightarrow{g} B$$

where C is some other object in $\mathcal{C}$

If $\mathcal{C}$ has pullbacks, we can compose spans, so given a span from A to B and B to C , we can take a pullback and obtain a span from A to C .



Maps between spans are then functions $f : C \rightarrow C'$ which make the diagram commute.



The bicategory $\text{Span}(\mathcal{C})$ then consists of

- Objects of $\mathcal{C}$ as 0-cells
- Spans as 1-cells
- Maps between spans as 2-cells

▲

Using the bicategory $\text{Span}(\mathbf{Set})$ in Def. 2.7 for a monad, we get the following.

Example 2.10 ([10]).

- The 0-cell is a set X

- The endo-1-cell is a span S of the form

$$X \xleftarrow{\delta_0} Y \xrightarrow{\delta_1} X$$

- The unit 2-cell is the function

$$\eta : X \rightarrow Y, \quad \delta_0 \circ \eta(x) = \delta_1 \circ \eta(x) = x$$

- The multiplication 2-cell $SS \rightarrow S$ is the function

$$\mu : Y \times_X Y \rightarrow Y, \quad \delta_0 \circ \mu(y, y') = \delta_0(y), \quad \delta_1 \circ \mu(y, y') = \delta_1(y')$$

where $Y \times_X Y = \{(y, y') | \delta_1(y) = \delta_0(y')\}$.

which satisfy the axioms of

- Associativity

$$\mu(\mu(y, y'), y'') = \mu(y, \mu(y', y''))$$

- Unit

$$\mu(y, \eta \circ \delta_1(y)) = y = \mu(\eta \circ \delta_0(y), y)$$

▲

Here the monad of $\text{Span}(\mathbf{Set})$ can be described as a category where

- X is the set of objects
- Y is the set of arrows
- $\delta_0, \delta_1 : Y \rightarrow X$ give the domain and codomain of each arrow respectively.
- The function $\eta : X \rightarrow Y$ gives the identity arrows $\eta(x) = 1_x$ for each $x \in X$.
- $Y \times_X Y$ is the set of composable arrows, and $\mu : Y \times_X Y \rightarrow Y$ defines composition.
- The associative and unit axioms are defined in the monad as they should be in the category.

Then, we have that this specific category as being a special case of a monad on the specific bicategory of $\text{Span}(\mathbf{Set})$.

c ... , a monad is a monoid, ...

At the same time, the concept of a monoid on sets $(M, \cdot, 1_M)$ can be generalised to categories. From Def. 1.3, the monoid on $(\mathbf{Set}, \times, \{*\})$ gives rise to the "monoid-as-a-set" from Def. 2.1. Here we give definitions to aid in the understanding of a monoid in $\mathbf{Set}$.

Definition 2.11 ([4]). A strict SMC is an SMC where $\alpha = 1, \lambda = \rho$. ($\sigma \neq 1!$).

▲

Definition 2.12 ([4]). A monoid in a strict monoidal category $(\mathcal{C}, \otimes, I)$ is an object X with morphisms

$$\mu : X \otimes X \rightarrow X, \quad \eta : I \rightarrow X$$

satisfying:

- associativity

$$\begin{array}{ccc} X \otimes X \otimes X & \xrightarrow{X \otimes \mu} & X \otimes X \\ \mu \otimes X \downarrow & & \downarrow \mu \\ X \otimes X & \xrightarrow{\mu} & X \end{array}$$

- left unit

$$\begin{array}{ccc} X & \xrightarrow{\eta \otimes X} & X \otimes X \\ & \searrow 1_X & \downarrow \mu \\ & & X \end{array}$$

- right unit

$$\begin{array}{ccc} X & \xrightarrow{X \otimes \eta} & X \otimes X \\ & \searrow 1_X & \downarrow \mu \\ & & X \end{array}$$

▲

Example 2.13 ([4]). , where

- Objects are a set X
- Unit object is a one element set $\{*\}$
-

$$a \cdot b := \mu(a, b)$$

-

$$1 := \eta(*)$$

$$(X, \mu : X \times X \rightarrow X, \eta : \{*\} \rightarrow X)$$

$$a \cdot b := \mu(a, b)$$

$$1 := \eta(*)$$

▲

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